OS & SP

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SKILL – 17

1.Skill17.c

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GNU nano 8.7.1 skill17.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_JOBS 100
#define MAX_CMD 256

typedef struct {
    int job_id;
    pid_t pid;
    char command[MAX_CMD];
    int running;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;
int next_job_id = 1;

/* Add background job */
void add_job(pid_t pid, const char *command)
{
    if (job_count >= MAX_JOBS) {
        printf("Job table full!\n");
        return;
    }

    jobs[job_count].job_id = next_job_id++;
    jobs[job_count].pid = pid;

    strncpy(jobs[job_count].command,
            command, MAX_CMD - 1);

    jobs[job_count].command[MAX_CMD - 1] = '\0';
    jobs[job_count].running = 1;

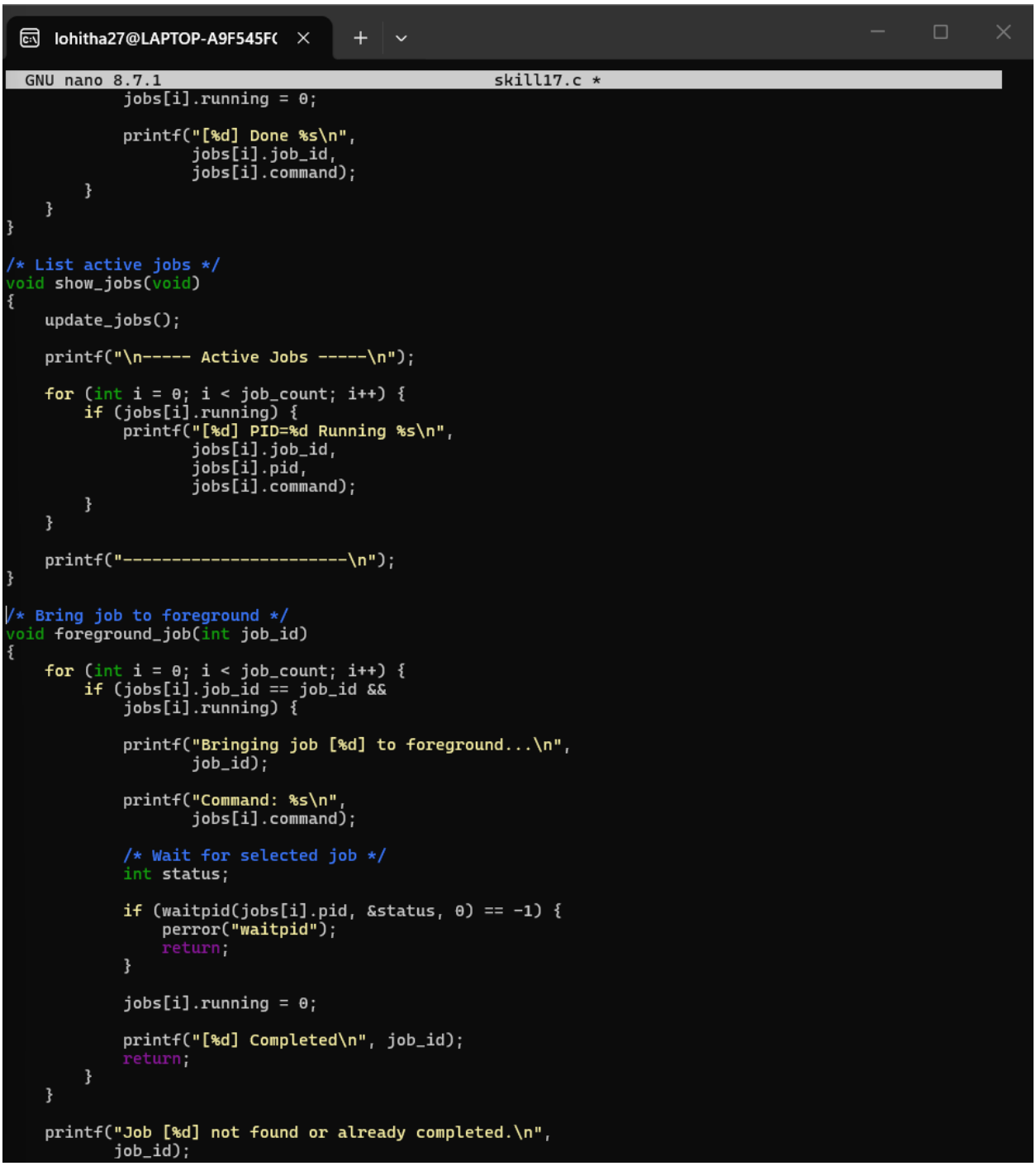
    printf("[%d] %d Running %s\n",
           jobs[job_count].job_id,
           jobs[job_count].pid,
           jobs[job_count].command);

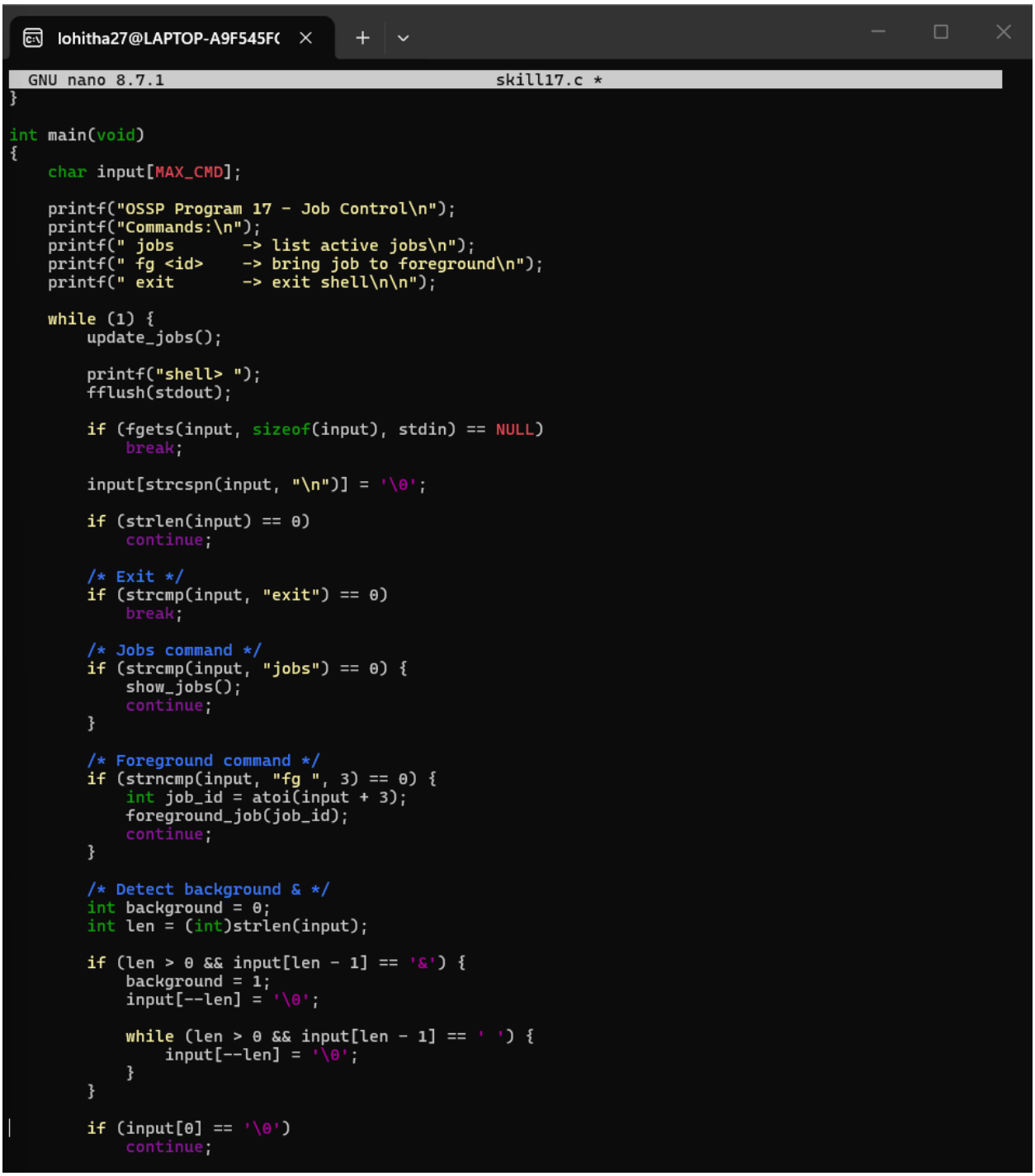
    job_count++;
}

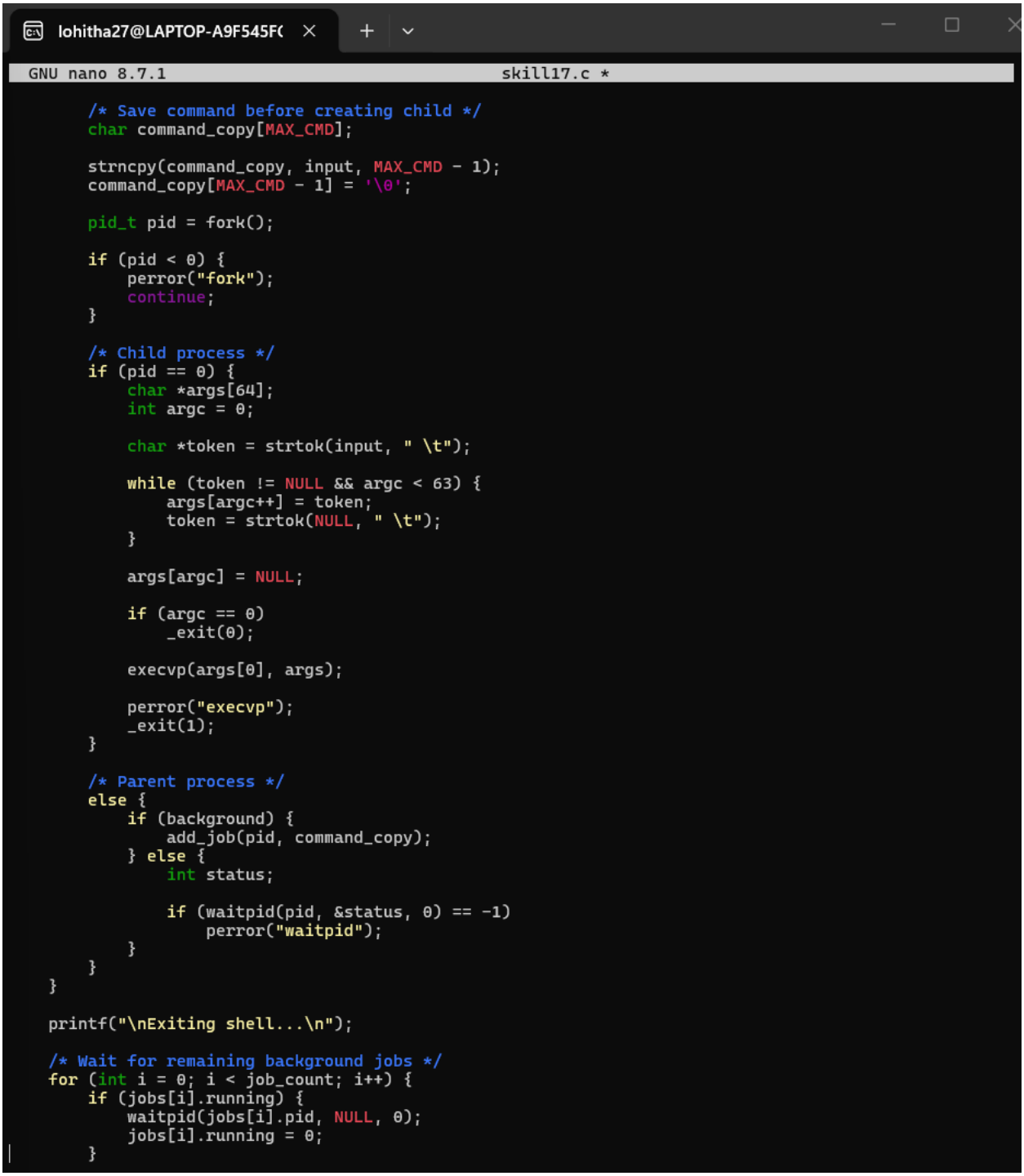
/* Update job status */
void update_jobs(void)
{
    for (int i = 0; i < job_count; i++) {
        if (!jobs[i].running)
            continue;

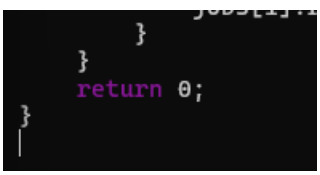
        int status;

        pid_t result = waitpid(
            jobs[i].pid, &status, WNOHANG
        );
```


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jobs[i].running = 0;
printf("[%d] Done %s\n",
jobs[i].job_id,
jobs[i].command);
}
}
}
/* List active jobs */
void show_jobs(void)
{
update_jobs();
printf("\n----- Active Jobs -----\n");
for (int i = 0; i < job_count; i++) {
if (jobs[i].running) {
printf("[%d] PID=%d Running %s\n",
jobs[i].job_id,
jobs[i].pid,
jobs[i].command);
}
}
printf("----------------------\n");
}
/* Bring job to foreground */
void foreground_job(int job_id)
{
for (int i = 0; i < job_count; i++) {
if (jobs[i].job_id == job_id &&
jobs[i].running) {
printf("Bringing job [%d] to foreground...\n",
job_id);
printf("Command: %s\n",
jobs[i].command);
/* Wait for selected job */
int status;
if (waitpid(jobs[i].pid, &status, 0) == -1) {
perror("waitpid");
return;
}
jobs[i].running = 0;
printf("[%d] Completed\n", job_id);
return;
}
}
printf("Job [%d] not found or already completed.\n",
job_id);


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}
int main(void)
{
char input[MAX_CMD];
printf("OSSP Program 17 - Job Control\n");
printf("Commands:\n");
printf(" jobs -> list active jobs\n");
printf(" fg <id> -> bring job to foreground\n");
printf(" exit -> exit shell\n\n");
while (1) {
update_jobs();
printf("shell> ");
fflush(stdout);
if (fgets(input, sizeof(input), stdin) == NULL)
break;
input[strcspn(input, "\n")] = '\0';
if (strlen(input) == 0)
continue;
/* Exit */
if (strcmp(input, "exit") == 0)
break;
/* Jobs command */
if (strcmp(input, "jobs") == 0) {
show_jobs();
continue;
}
/* Foreground command */
if (strncmp(input, "fg ", 3) == 0) {
int job_id = atoi(input + 3);
foreground_job(job_id);
continue;
}
/* Detect background & */
int background = 0;
int len = (int)strlen(input);
if (len > 0 && input[len - 1] == '&') {
background = 1;
input[--len] = '\0';
while (len > 0 && input[len - 1] == ' ') {
input[--len] = '\0';
}
}
if (input[0] == '\0')
continue;


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/* Save command before creating child */
char command_copy[MAX_CMD];
strncpy(command_copy, input, MAX_CMD - 1);
command_copy[MAX_CMD - 1] = '\0';
pid_t pid = fork();
if (pid < 0) {
perror("fork");
continue;
}
/* Child process */
if (pid == 0) {
char *args[64];
int argc = 0;
char *token = strtok(input, " \t");
while (token != NULL && argc < 63) {
args[argc++] = token;
token = strtok(NULL, " \t");
}
args[argc] = NULL;
if (argc == 0)
_exit(0);
execvp(args[0], args);
perror("execvp");
_exit(1);
}
/* Parent process */
else {
if (background) {
add_job(pid, command_copy);
} else {
int status;
if (waitpid(pid, &status, 0) == -1)
perror("waitpid");
}
}
}
printf("\nExiting shell...\n");
/* Wait for remaining background jobs */
for (int i = 0; i < job_count; i++) {
if (jobs[i].running) {
waitpid(jobs[i].pid, NULL, 0);
jobs[i].running = 0;
}


}
}
return 0;
}

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill17.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill17.c -o skill17
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill17
OSSP Program 17 - Job Control
Commands:
 jobs       -> list active jobs
 fg <id>    -> bring job to foreground
 exit       -> exit shell

shell> jobs

----- Active Jobs -----
-----------------------
shell> gf 1
execvp: No such file or directory
shell> exit

Exiting shell...
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```